

OMAR FOUDA

MACHINE LEARNING ENGINEER | APPLIED AI, RAG & COMPUTER VISION

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PROFESSIONAL SUMMARY

Recent Information Systems graduate focused on machine learning and applied AI, with hands-on experience delivering an AI assistant for a live education product. Built production-oriented systems in RAG, semantic search, CV matching, computer vision, and prediction APIs using Python, FastAPI, vector databases, Docker, and modern LLM tooling.

EXPERIENCE

AI Engineer | Nabra

Paid product engagement | Live product: nabrahq.com

- Adapted an Arabic-first educational RAG backend into the AI assistant powering Nabra's live platform for quiz, summary, and learning-content generation.
- Designed AI-driven analytics that surface content-improvement suggestions and flag students at risk, translating model outputs into actionable educator workflows.
- Shipped features used by real subscribed customers while collaborating directly with the startup's co-founders.

SELECTED PROJECTS

Educational RAG Backend

Python | FastAPI | ChromaDB | OpenAI/Gemini | OCR | Whisper | FFmpeg | Docker | [GitHub](#)

- Built a multimodal RAG backend that turns documents, images, audio, and video into searchable educational content through OCR, transcription, and media-processing pipelines.
- Implemented metadata-aware retrieval across 5 context dimensions - subject, grade, term, book, and course - to improve answer relevance for Arabic learning content.
- Created Dockerized local and low-cost cloud deployment profiles with switchable providers for embeddings, OCR, and transcription.

Jobify AI Matching Backend | Graduation Project

Python | FastAPI | FAISS | Sentence Transformers | Gemini | SQL Server | SQLite | [GitHub](#)

- Built an AI backend for job recommendations, CV analysis, and career chat, connected to a hosted SQL Server application database.
- Combined FAISS semantic search with SQL fallback and active CV retrieval to match candidates with jobs and expose match scores, skill gaps, and improvement advice.
- Added PDF/DOCX parsing, embedding synchronization, health checks, and admin endpoints to keep job and resume indexes current.

Diabetes Prediction Public API

Python | FastAPI | scikit-learn | REST | ngrok | [GitHub](#)

- Deployed a trained scikit-learn classifier behind a public REST API with a validated 8-feature JSON contract, human-readable responses, and automated endpoint tests.

TECHNICAL SKILLS

Core: Python, SQL, NumPy, Pandas, scikit-learn, TensorFlow/Keras, PyTorch

Applied AI: RAG, semantic search, Sentence Transformers, OpenAI API, Gemini API, Anthropic API, MCP, ChromaDB, FAISS, OpenCV, Tesseract OCR, Whisper

Backend & Delivery: FastAPI, REST APIs, WebSockets, Docker, Git/GitHub, SQL Server, PostgreSQL, SQLite

EDUCATION & PROFESSIONAL DEVELOPMENT

B.Sc. Computers and Information (Information Systems)

Menoufia University | 2026

AWS Training & Certification

Fundamentals of Machine Learning and AI | Aug 2026

Selected Anthropic Training (2026): Claude API | Model Context Protocol (Introduction & Advanced) | Claude Code (101 & In Action) | Agent Skills | Amazon Bedrock & Google Vertex AI

Additional Certifications: Kaggle Computer Vision & Intro to Deep Learning (Nov 2025) | ITI Full Stack Web Development with Python, 160h (Jul-Aug 2025) | Manara Responsible AI & Ethics (Jun 2024)

LANGUAGES

Arabic - Native | English - Professional working proficiency